



DATA INTEGRATION AND ENRICHMENT

Make your data usable

1

Schema matching

2

Instance level matching

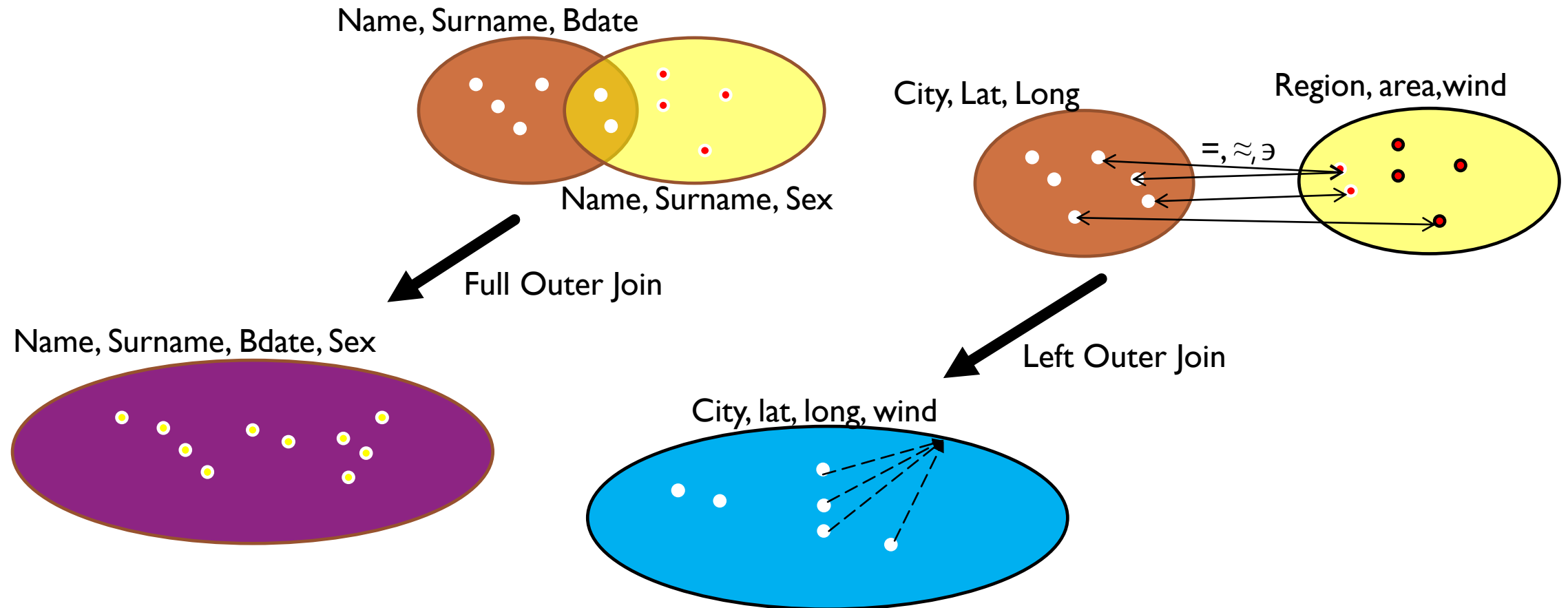
3

Deduplication

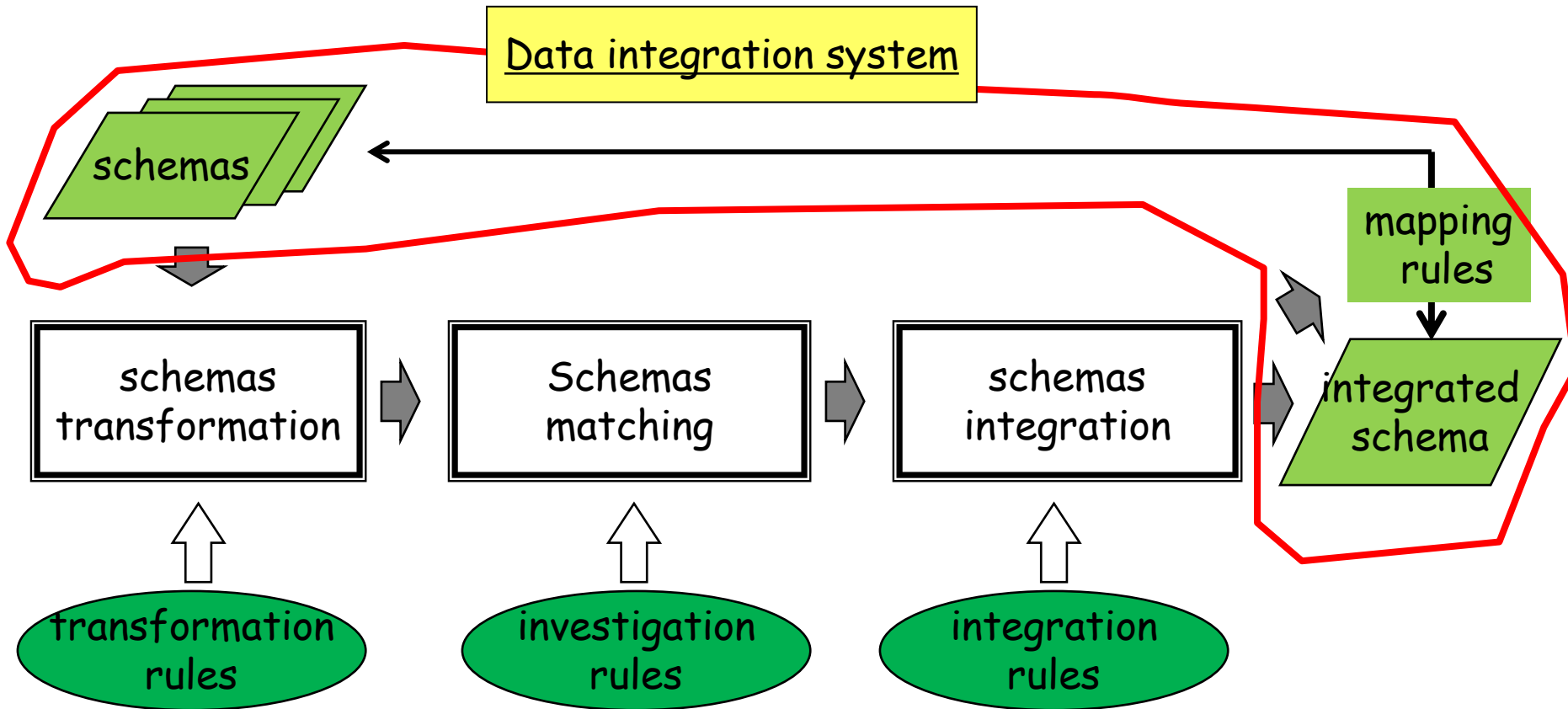
4

Record linkage

DATA INTEGRATION VS ENRICHMENT

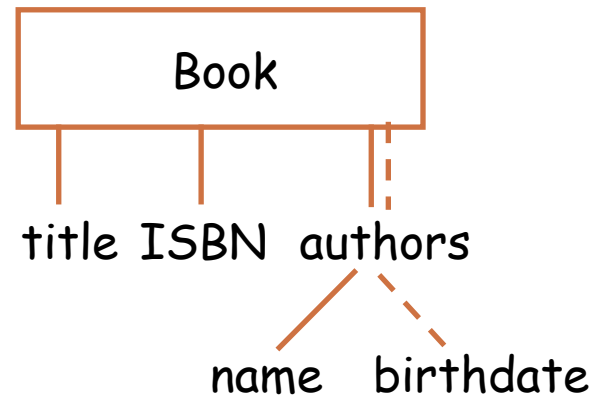


A GENERIC FRAMEWORK FOR INTEGRATION

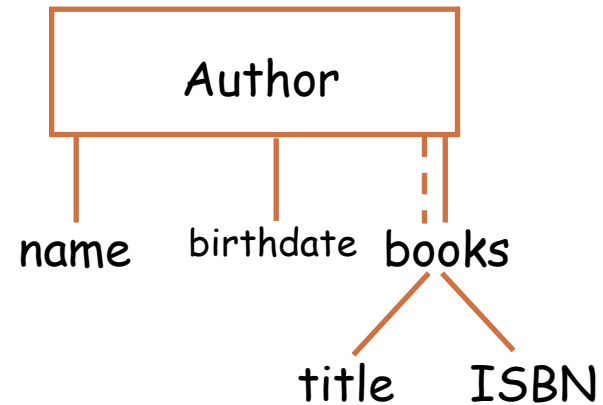


LIBRARY EXAMPLE (HOMOGENEOUS MODELS)

Schema 1

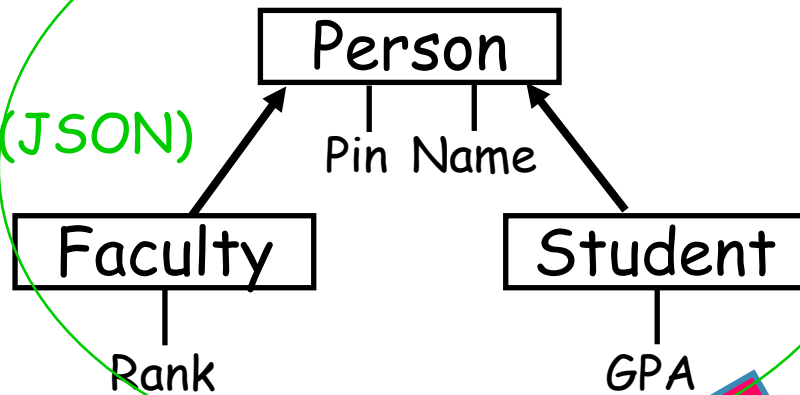


Schema 2



THESIS EXAMPLE (HETEROGENEOUS M)

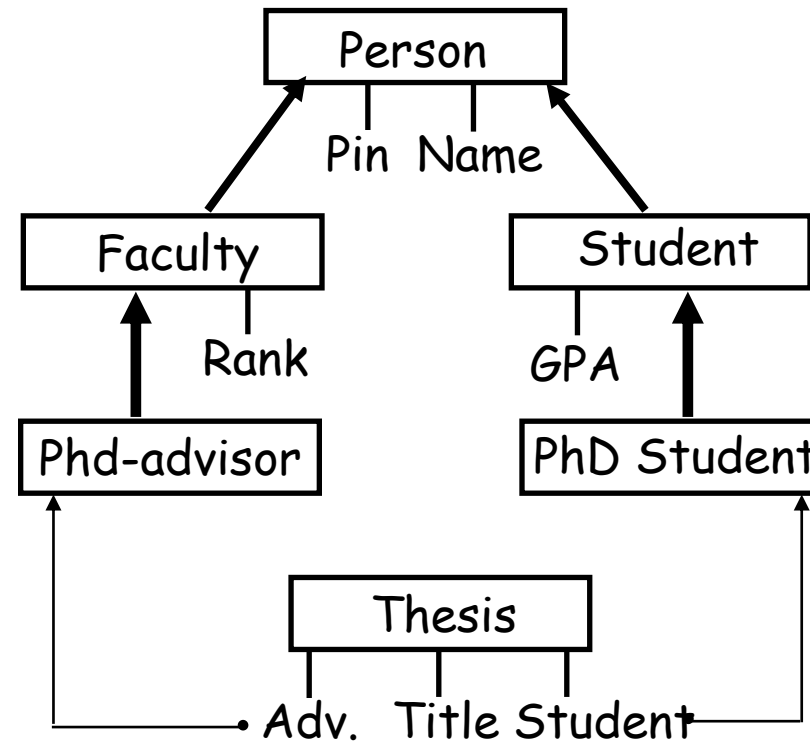
Schema S1 (JSON)



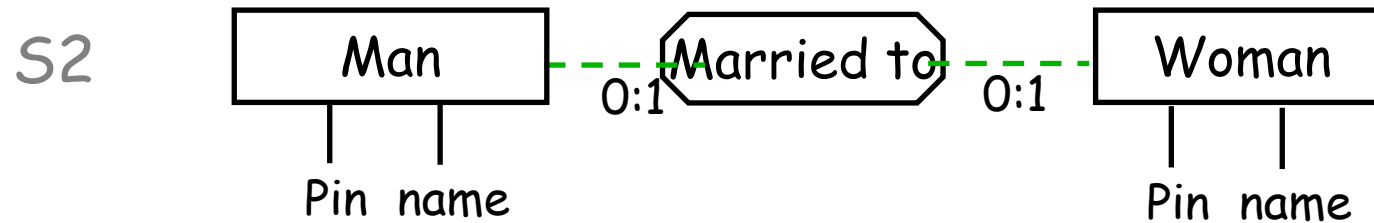
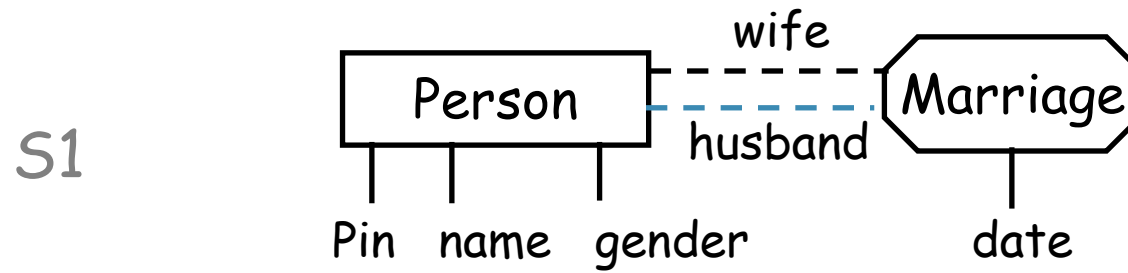
Schema S2 (relational)

Thesis (Phd-advisor, Phd-student, title)

The integrated schema (JSON)



MARRIAGE EXAMPLE



SCHEMA LEVEL HETEROGENEITIES

- NB heterogeneity, conflict and correspondence are synonyms in the following

Are of two types

- Name heterogeneities
- Type heterogeneities

Besides name and type heterogeneities we have to consider in the more general context also

- Model heterogeneities

NAME HETEROGENEITIES (CONFLICTS/ CORRESPONDENCES)

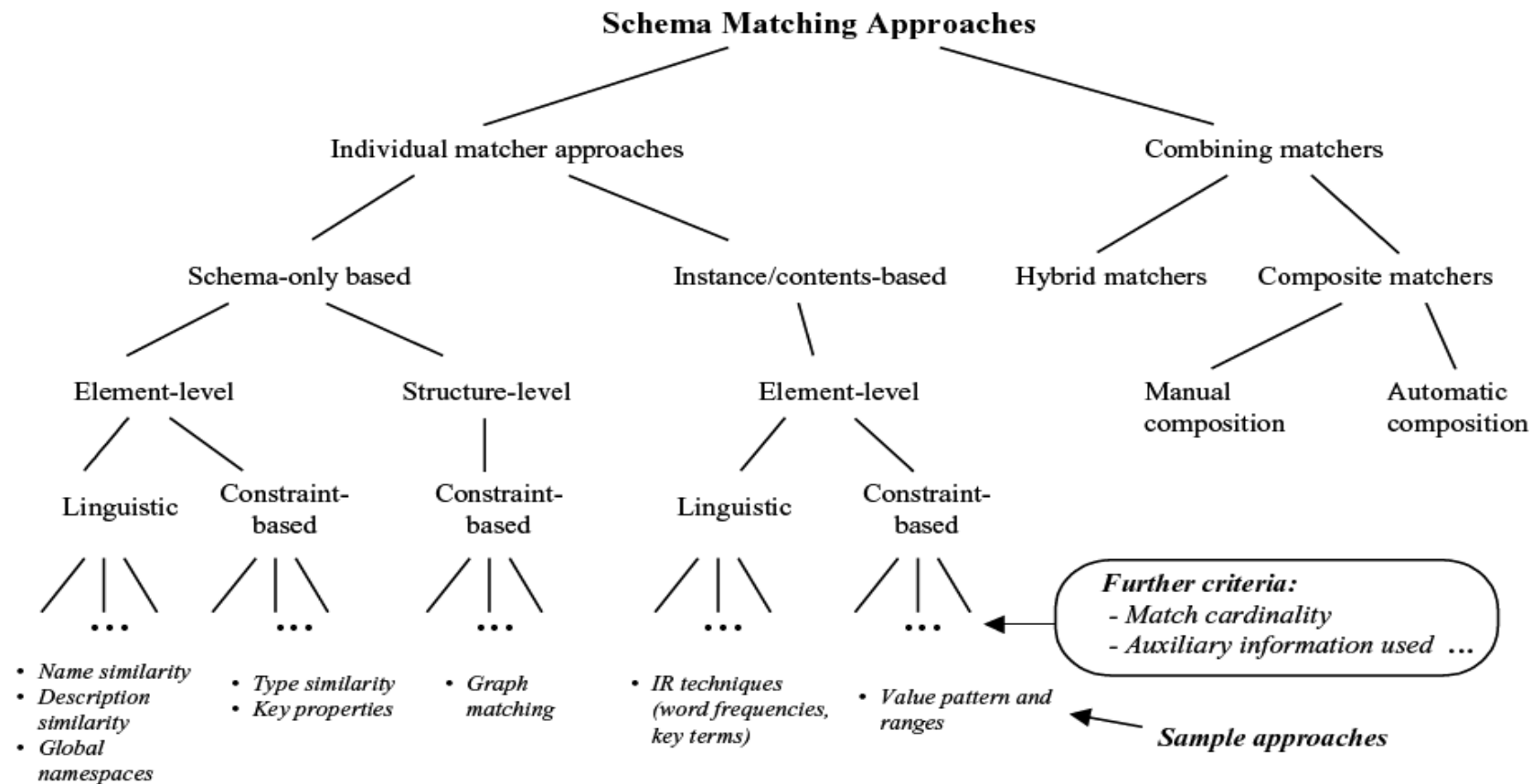
- Synonyms – Different names for the same concepts
 - employee, clerk
 - exam, course
 - code, num
- Homonyms – Same name for different concepts
 - City as City of birth in one schema, as City of Residence in another schema
- Hyperonymies – Two concepts related by an IS-A relationship

TYPE HETEROGENEITIES

The same concepts is represented with different conceptual structures in two schemas

- Different definition domains for the same attribute in two schemas
- Attribute in one schema and derived value in another schema
- Attribute in one schema and entity in another schema
- Attribute in one schema and generalization hierarchy in another schema
- Entity in one schema and relationship in another schema
- Different abstraction levels for the same concept in two schemas: e.g. two entities with homonym names related by an IS-A hierarchy in two schemas
- Different granularities in the definition domains
- Different cardinalities in the same relationships
- Key conflicts

CLASSIFICATION OF TECHNIQUES (APPROACHES)



Schema matching

PHASES OF THE METHODOLOGY, INPUTS, OUTPUTS, AND METHODS USED

01. Schema transformation (or Pre-integration)

Input: n source schemas

Output: n source schemas homogeneized

Methods used: Model transformation + Reverse engineering

2. Correspondences investigation

Input: n source schemas

Output: n source schemas + correspondences

Method used: techniques to discover correspondences

3. Schemas integration and mapping generation

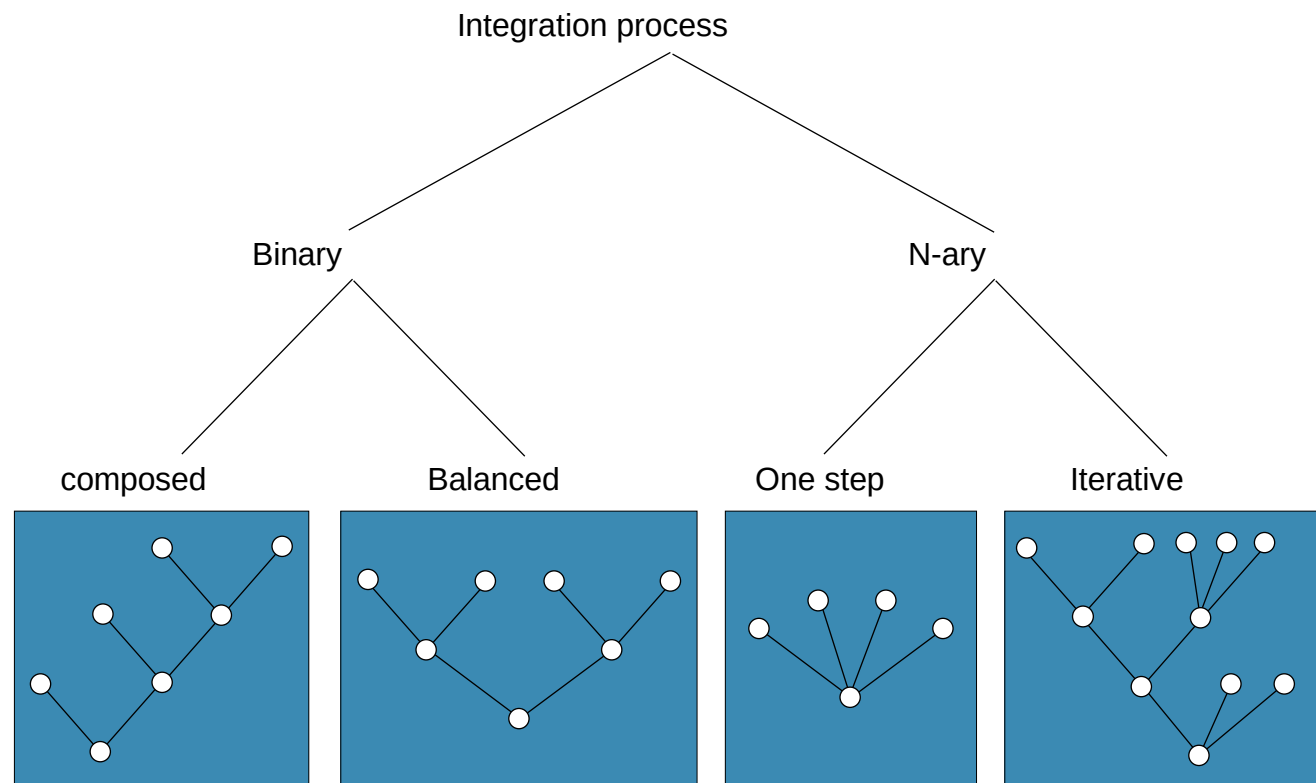
Input: n source schemas + correspondences

Output: integrated schema + mapping rules btw the integrated schema and input source schemas

Method used: New classification of conflicts + Conflict resolution transformations

PRE INTEGRATION

■ Strategies



PHASES OF THE METHODOLOGY, INPUTS, OUTPUTS, AND METHODS USED

0. 1. Schema transformation (or Pre-integration)

Input: n source schemas

Output: n source schemas homogeneized

Methods used: Model transformation + Reverse engineering

2. Schema matching (Correspondences investigation)

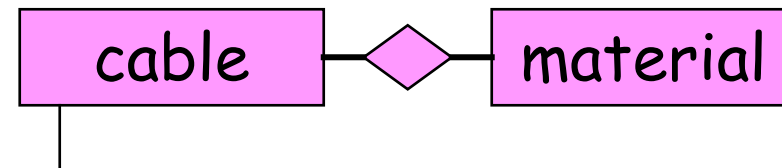
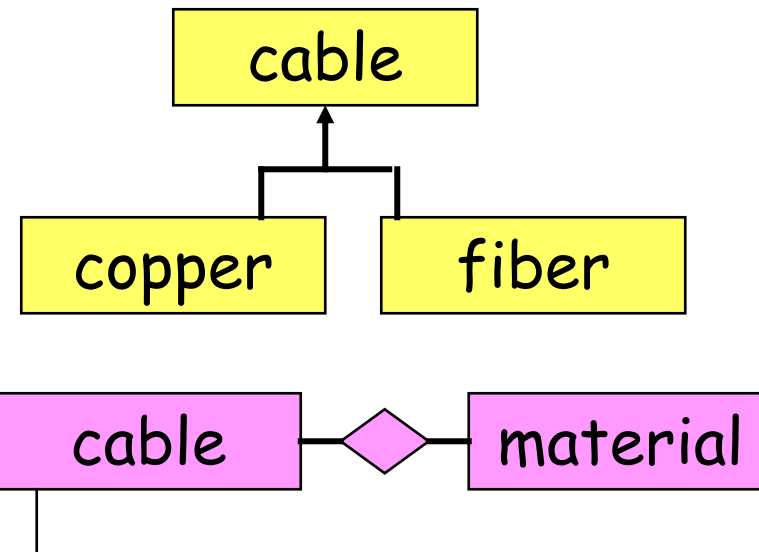
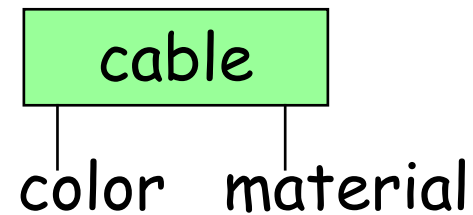
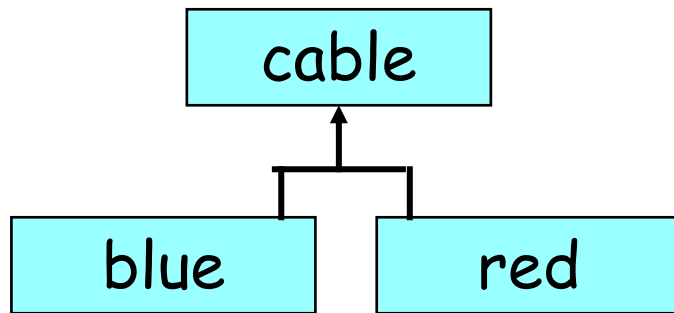
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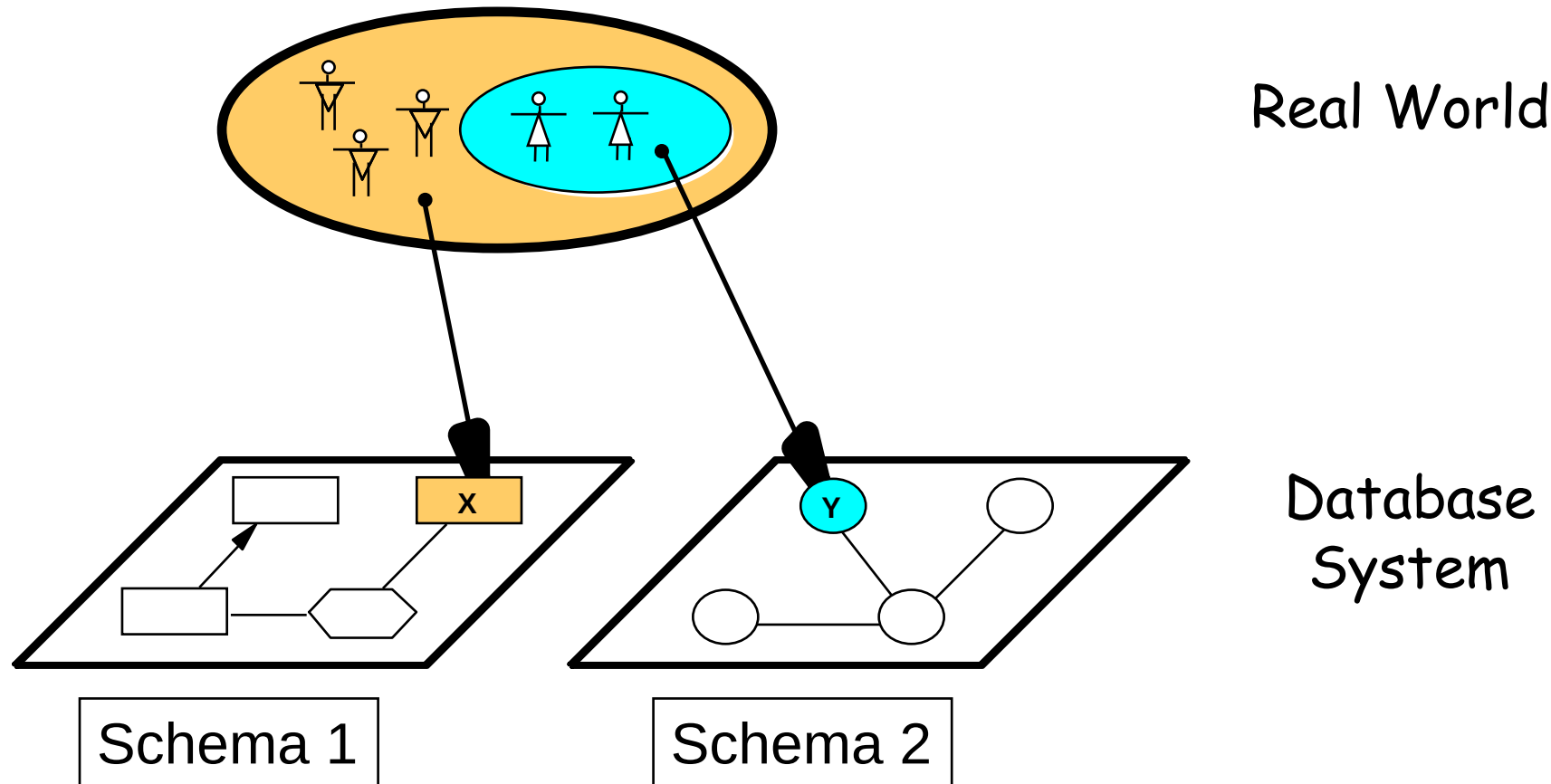
Method used: techniques to discover correspondences

SEMANTIC RELATIVISM

- Information representation depends on perception

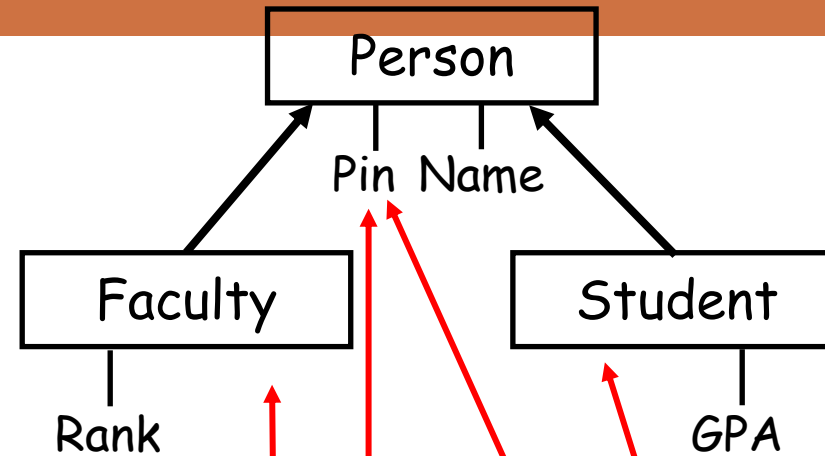


SEMANTICS OF CORRESPONDENCES



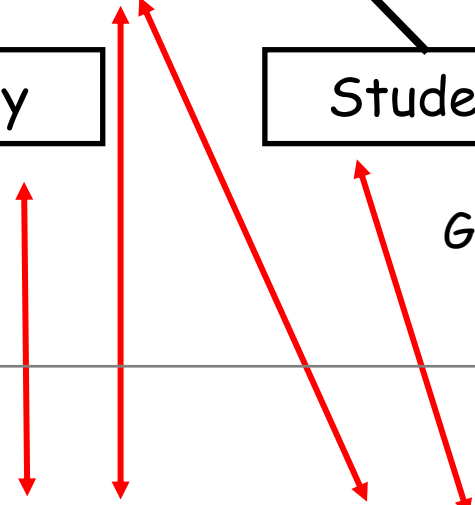
EXAMPLES OF CORRESPONDENCES

Schema S1 (JSON)



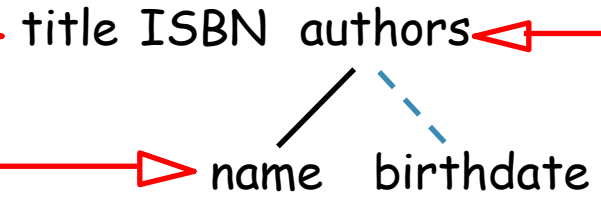
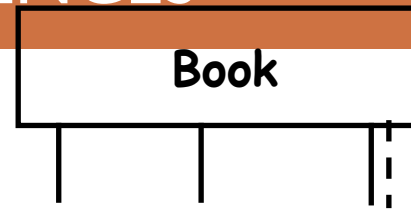
Schema S2 (relational)

Thesis (Phd-advisor, Phd-student, title)

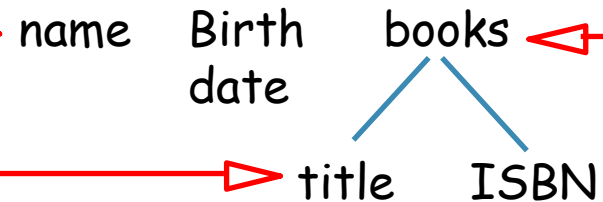
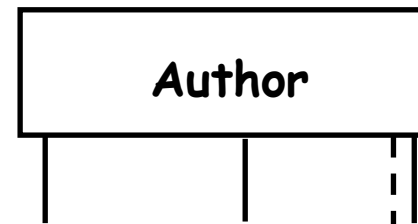


SOME LIBRARY CORRESPONDENCES

- Schema 1



- Schema 2



MATCH SELECTOR

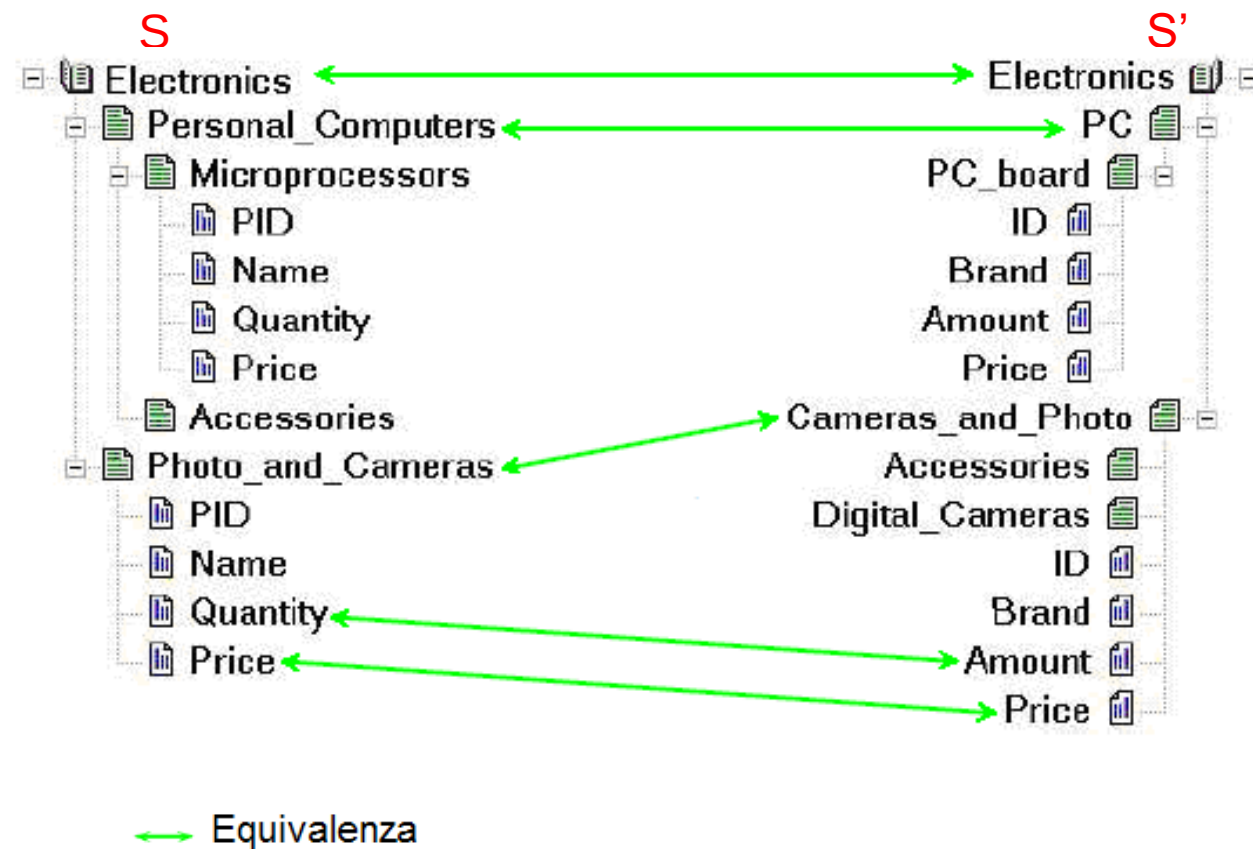
S

S'

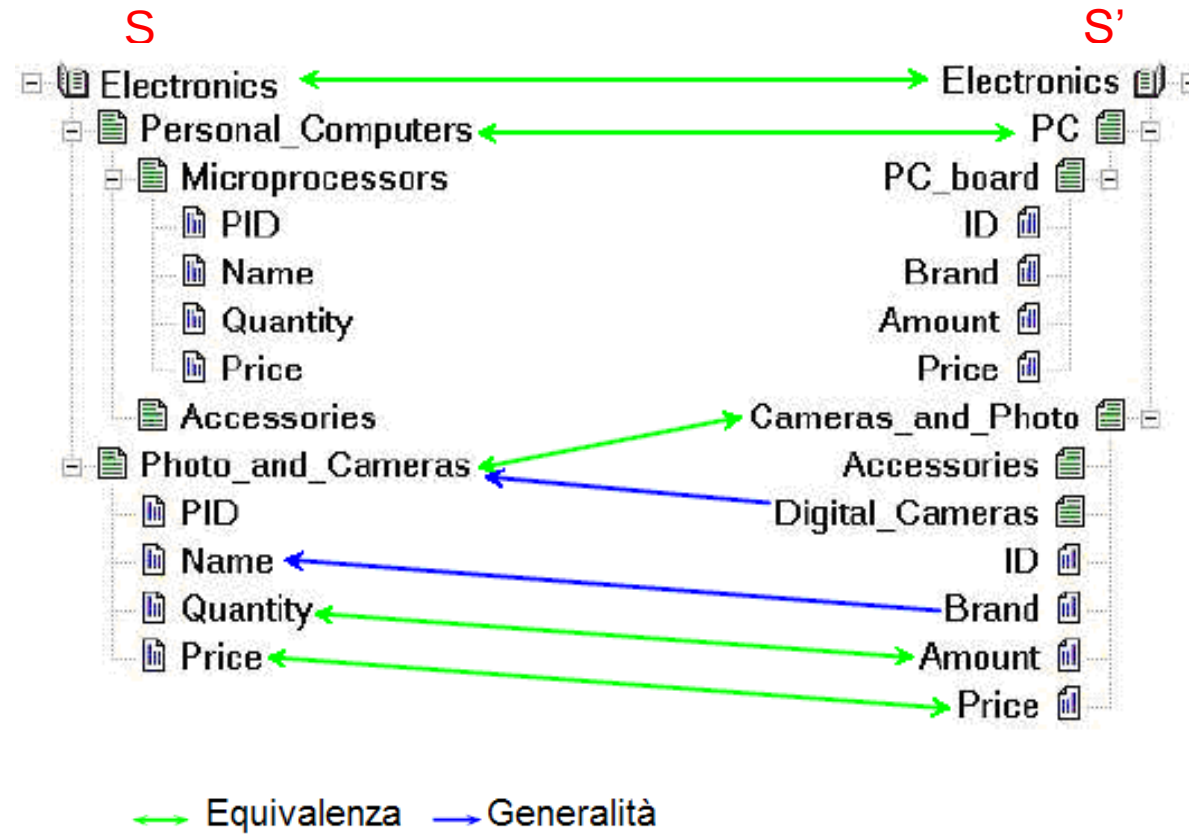
- What are the candidate's identifiers?



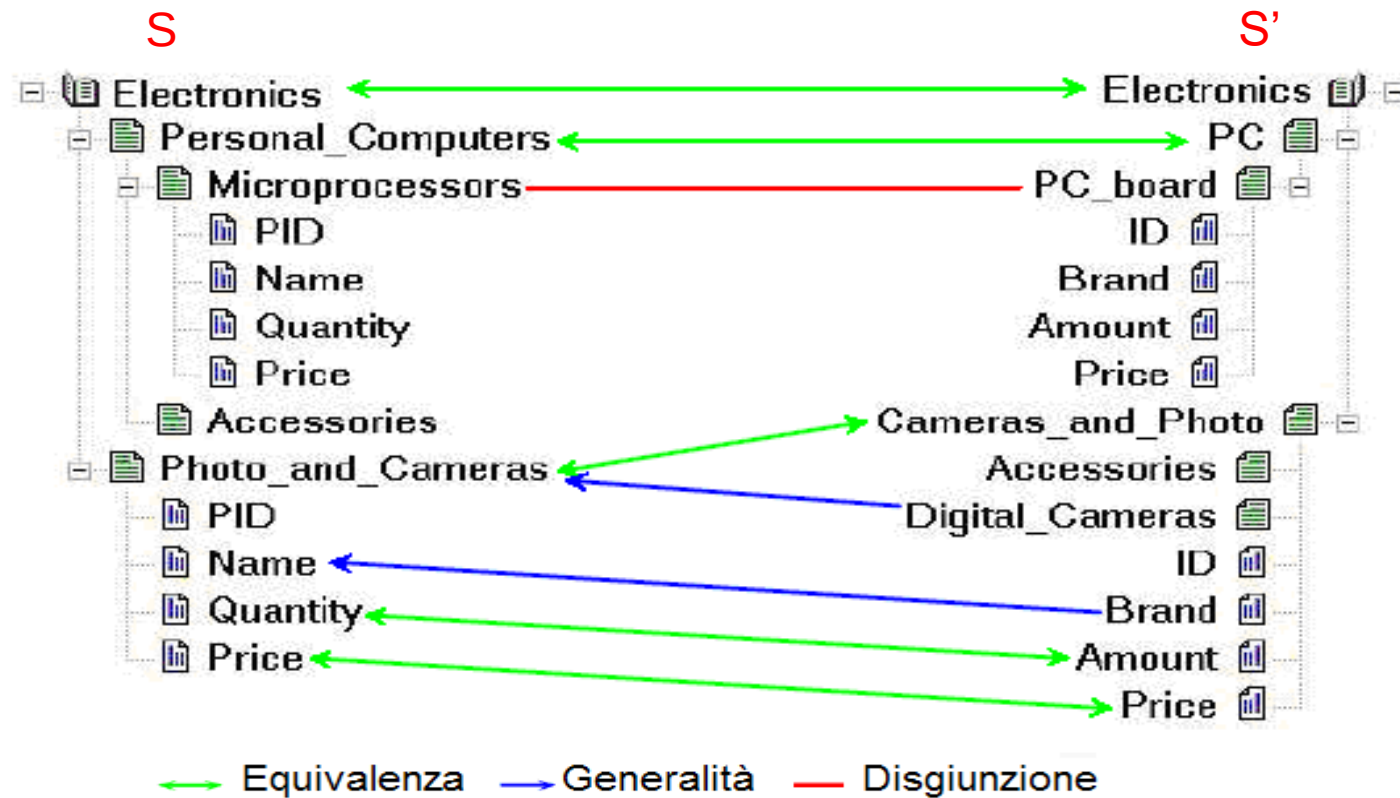
EQUIVALENCE



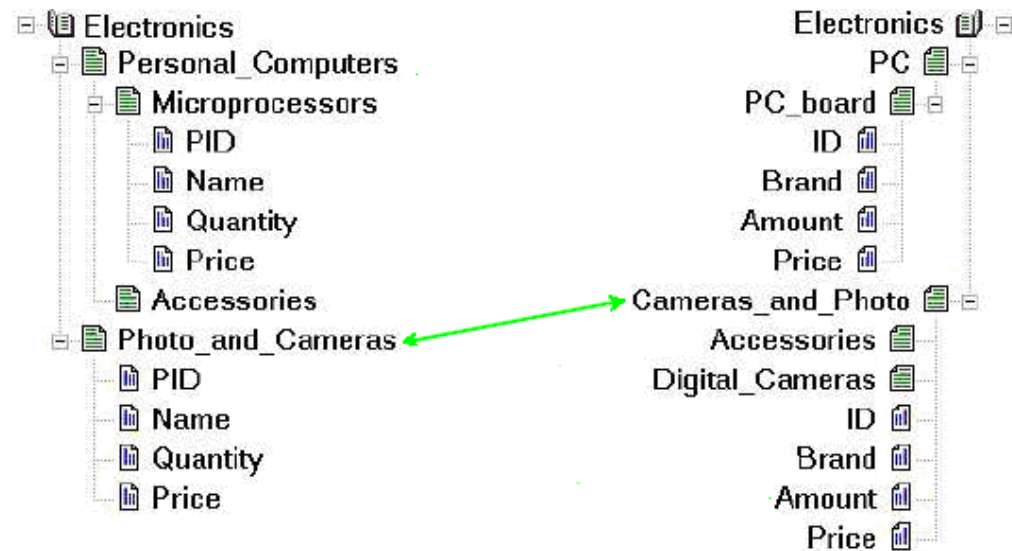
IS-A



DISJUNCTION



SIMILARITY ESTIMATOR AND MATCH SELECTOR



PHASES OF THE METHODOLOGY, INPUTS, OUTPUTS, AND METHODS USED

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2. Correspondences investigation

Input: n source schemas

Output: n source schemas + correspondences

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Input: n source schemas + correspondences

Output: integrated schema + mapping rules btw the integrated schema and input source schemas

Method used: New classification of conflicts + Conflict resolution transformations

CONFLICTS REVISITED

- Find matching correspondences could be performed through the use of a rich language for expressing correspondences, has fixed the types of conflicts expressed on the schemas. Let us first provide
- the new classification of types of conflicts, and
- corresponding conflict resolution transformations (also called integration rules)

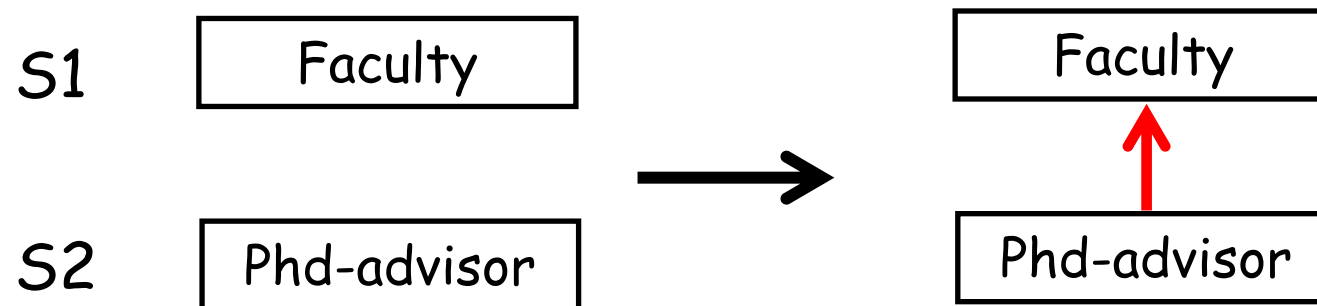
I. CLASSIFICATION CONFLICTS

Corresponding elements describe different sets of real world objects

Correspondence: S1.Faculty CONTAINS S2.PhD-advisor

- Conflict Resolution:

- Introduction of a Generalization/Specialization hierarchy



2. DESCRIPTIVE CONFLICTS - I

- Corresponding types have different properties, or corresponding properties are described in different ways
- Object / Entity / Relationship type:
 - naming conflicts :
 - synonyms Node , Extremity
 - homonyms Highway (EU) , Highway (USA)
 - composition conflicts : different attributes and methods
 - Employee (E# , name , address)
 - Employee (E# , position , salary , department)

2. DESCRIPTIVE CONFLICTS - 2

- Solution depends upon:
 - the type of the descriptive conflict
 - the type of classification conflict

S1 : Employee (E# , name , address)

S2 : Employee (E# , position , salary)

S1.Employee EQUAL S2.Employee =>

IS. Employee (E# , name , address , position , salary)

S1.Employee CONTAINS S2.Employee =>

IS. Employee (E# , name , address , [position , salary])

3. STRUCTURAL CONFLICTS

- Different schema element types, e.g.: class, attribute, relationship
- Library example :
 - S1 : Book is an Entity
 - S2 : Books is an attribute of Author
- Spatial example :
 - S1 : Tunnel is an attribute of RoadSection
 - S2 : Tunnel is an Entity
- Conflict resolution :
 - Choose the less constraining structure
 - IS: Book, Tunnel are Entities

4. FRAGMENTATION CONFLICTS (IN TRADITIONAL DATABASES)

- The same phenomenon of the real world is perceived as a single object in one database and as several objects in the other
- Example
 - DB1: Soccer Team
 - DB2: Soccer Player
- Solution
 - aggregation relationship

S1 Team

S2 Player



S1 Team

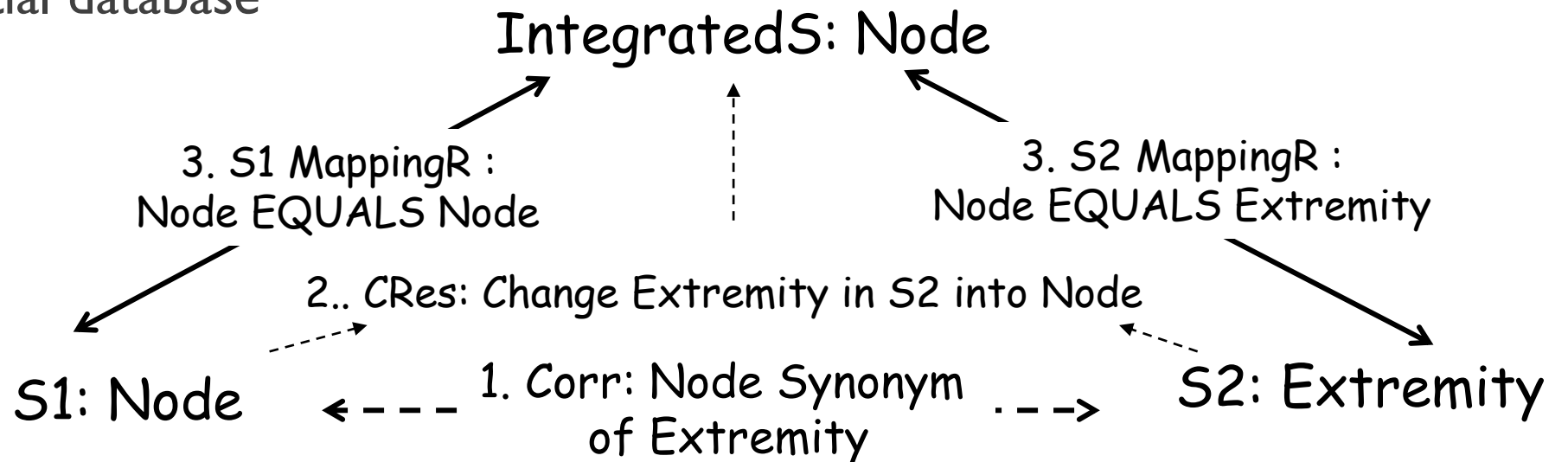


S2 Player

Aggregator operator, usually not present natively in conceptual models, has to be simulated

DEFINITION OF MAPPING RULES

- Defined between the integrated schemas and the source schemas
- Can be found for the source schema S looking at correspondences and conflict resolution transformations involved adopted for S in the integration process
- Example in a spatial database



INSTANCE-LEVEL CONFLICT RESOLUTION

- Same data in different sources have different values
- Fundamental problem in both virtual and materialized data integration

INSTANCE LEVEL CONFLICTS - I

- Given two sources S_1 and S_2 , let A_{1k} and A_{2k} represent the same attributes of a real world object represented as t_1 in S_1 and as t_2 in S_2 :
 - **Attribute Conflicts:** An attribute conflict occurs if $t_1.A_{1k} \neq t_2.A_{2k}$
 - **Key Conflicts** (also called entity or tuple conflicts): A_{1k} is primary key for t_1 and A_{2k} is primary key for t_2 . A Key conflict occurs iff $t_1.A_{1k} \neq t_2.A_{2k}$ and $t_1.A_{1i} = t_2.A_{2i}$ for all $i \neq k$

TYPES OF INSTANCE LEVEL CONFLICTS

EmployeeID	Name	Surname	Salary	Email
arpa78	John	Smith	2000	smith@abc.it
eugi98	Edward	Monroe	1500	monroe@abc.it
ghjk09	Anthony	Wite	1250	white@abc.it
treg23	Marianne	Collins	1150	collins@abc.it

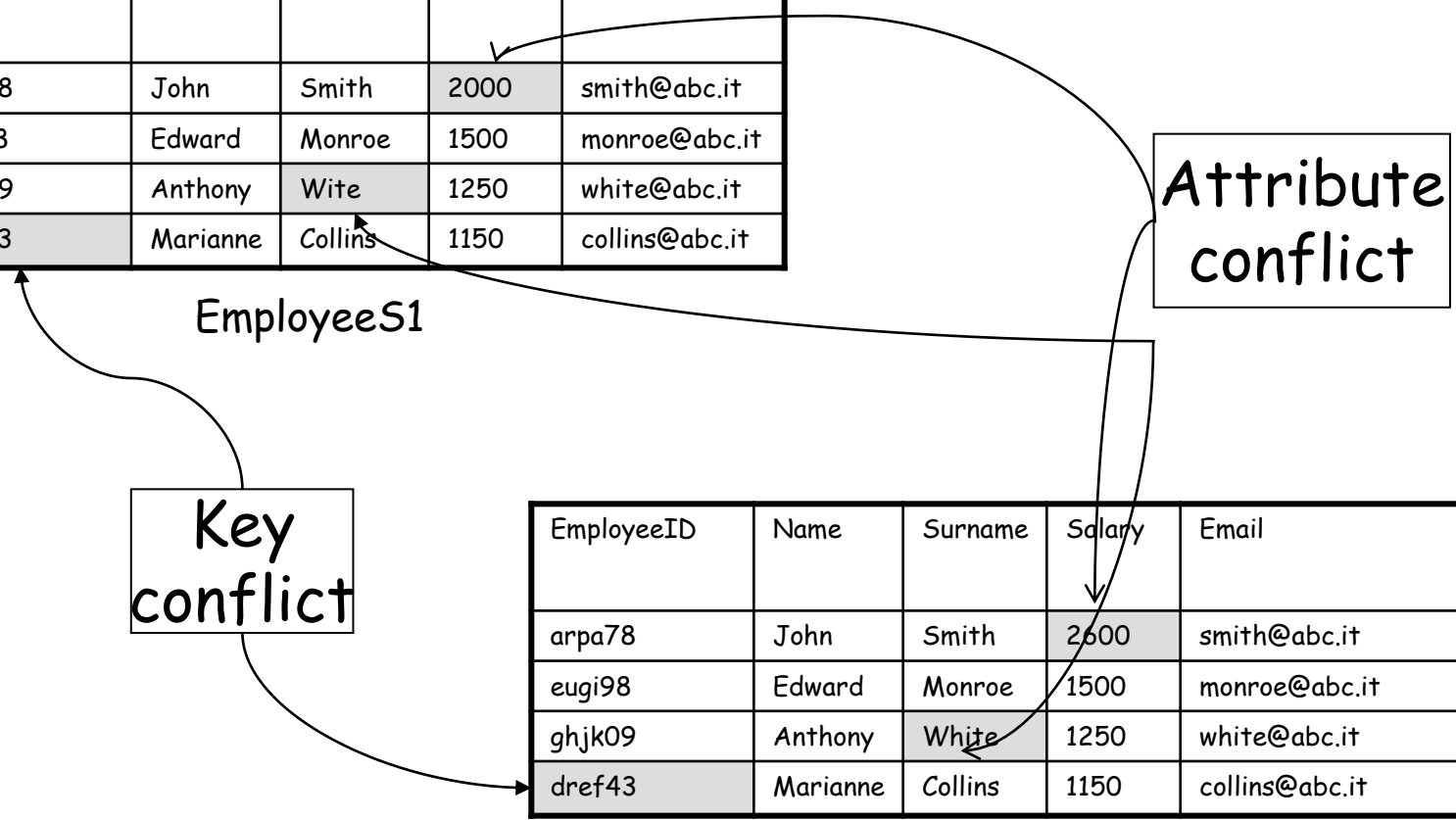
EmployeeS1

Key
conflict

EmployeeID	Name	Surname	Salary	Email
arpa78	John	Smith	2000	smith@abc.it
eugi98	Edward	Monroe	1500	monroe@abc.it
ghjk09	Anthony	White	1250	white@abc.it
dref43	Marianne	Collins	1150	collins@abc.it

EmployeeS2

Attribute
conflict



INSTANCE LEVEL CONFLICTS

- Techniques that deal with instance-level conflicts can be applied in two different phases of the life cycle of a data integration system, namely, at *design time* and at *query time*.
- In both cases, the actual conflicts occur at query time; however, the design time approaches decide the strategy to follow for fixing conflicts before queries are processed, i.e., at the design stage of the data integration system.
- The techniques operating at query time incorporate the specification of the strategy to follow within query formulation.

RESOLUTION FUNCTION

A *resolution function* takes two (or more) conflicting values of an attribute as input and outputs a value that must be returned as the result to the posed query.

Common resolution functions are MIN and MAX. For non numerical attributes, further resolution functions can be identified, such as CONCAT.

POSSIBLE RESOLUTION FUNCTIONS

Function	Attribute Type	Description
COUNT	any	Counts number of conflicting values
MIN	any	Minimum value
MAX	any	Maximum value
RANDOM	any	Random non null value
CHOOSE(Source)	any	Chooses most reliable source for the particular attribute
MAXIQ	any	Value of highest information quality
GROUP	any	Groups all conflicting values
SUM	numerical	Sums all values
MEDIAN	numerical	Median value, namely having the same number of higher and lower values
AVG	numerical	Arithmetic mean of all values
VAR	numerical	Variance of values
STDDEV	numerical	Standard Deviation of values
SHORTEST	non-numerical	Minimum length value, ignoring spaces
LONGEST	non-numerical	Maximum length value, ignoring spaces
CONCAT	non-numerical	Concatenation of values
ANNCONCAT	non-numerical	Annotated concatenation of values, whose purpose is to specify the source, before the actual returned value

DATA INTEGRATION MORE IN DEPTH

Data Integration has two shapes:

- Deduplication → Integration of data inside one table
- Integration of data in two or more tables



DEDUPLICATION



DEDUPLICATION:

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	null
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Are there records in the Table that refer to the same person in the real world?

DEDUPLICATION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	null
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

?

Rule of thumb: records refer to the the same real world object if they differ only in one letter in Name & Surname

DEDUPLICATION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Rule of thumb: the same if records differ only in one letter in Name & Surname and Name is the same, although in different languages

DEDUPLICATION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Question: Which is the correct name?

DATA QUALITY ASSESSMENT

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Carla
Carlo
Cassio
Caterina
Catullo
...
Mario



Carla
Carlo
Cassio
Caterina
Catullo
...
Mario

Compare Name with italian names
and choose the one at distance 0

DATA QUALITY ASSESSMENT → DATA FUSION

Name	Surname	Place of birth	Country
Erle	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Compare Name with italian names and choose the one at distance 0

DATA FUSION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Pesaro
Pescara
Pescina
...



Pesaro
Pescara
Pescina
...

Rule of thumb: differ only
in one letter in Name & Surname
Question: Which is the correct Place of Birth?

DATA FUSION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Rule of thumb: differ only in one letter in Name & Surname

Choose the Place of Birth that is at distance 0 from Places in the directory of Italian Municipalities

DATA FUSION

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pscar	Italy

Rule of thumb: differ only in one letter in Name & Surname

Choose the Place of Birth that is at distance 0 from Places in the directory of Italian Municipalities

FINAL FUSION AND PROVENANCE

Name	Surname	Place of birth	Country
Crlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy
Carlo	Batini	Pescara	Italy

Name	Surname	Place of birth	Country
Carlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy

DEDUPLICATION

Name	Surname	Place of birth	Country
Carlo	Batini	Pescara	Italy
Karl	Batini	Pescara	Italy
Giulio	Batini	null	Italy

Final result of deduplication and fusion
is it correct?

DEDUPLICATION

FINAL RESULT OF DEDUPLICATION AND FUSION

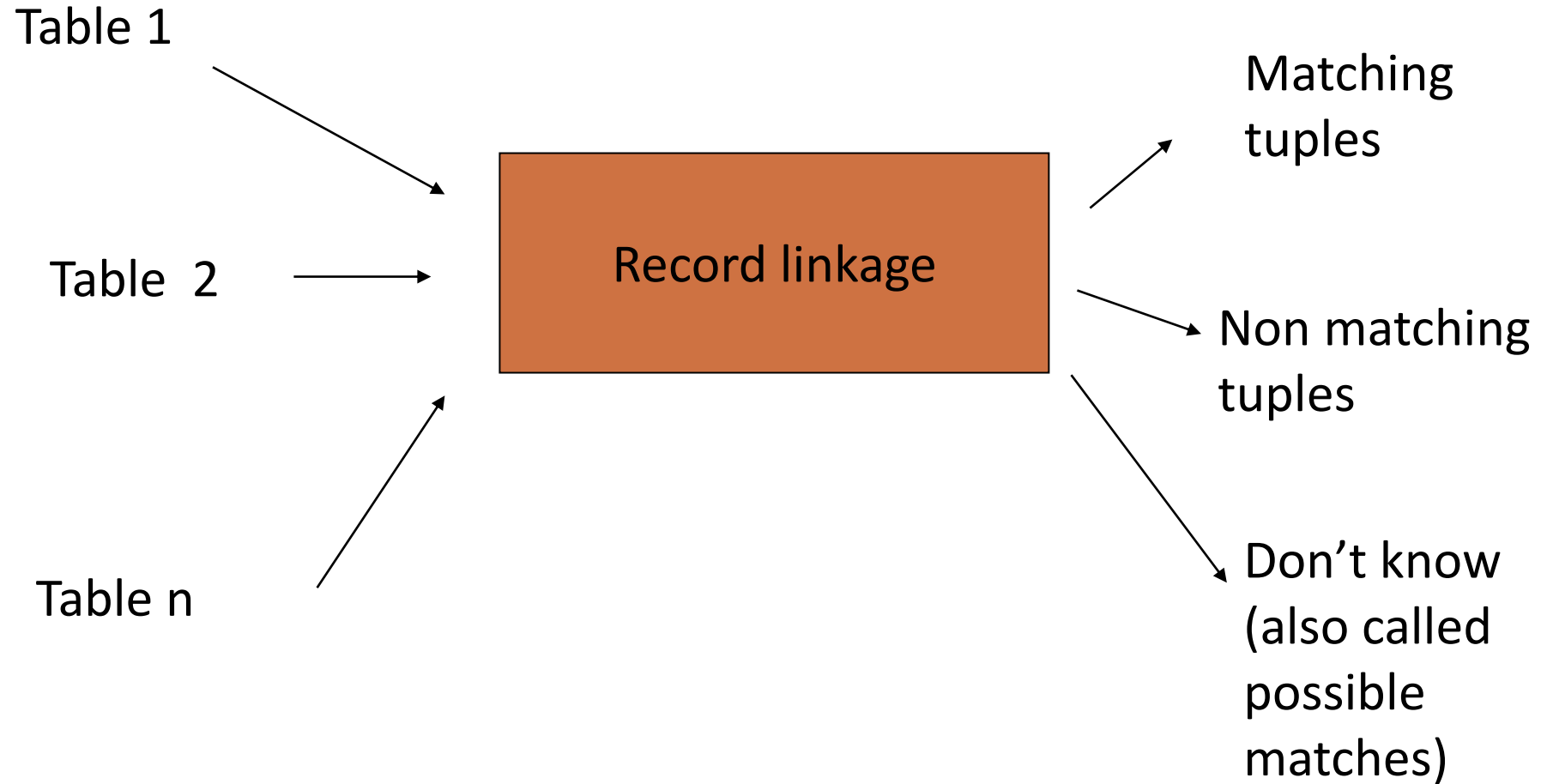
Name	Surname	Place of birth	Country
Carlo	Batini	Pescara	Italy
Giulio	Batini	null	Italy



DATA INTEGRATION OF TWO TABLES



RECORD LINKAGE: INPUT OUTPUT DESCRIPTION



RECORD LINKAGE: INPUT OUTPUT DESCRIPTION

Table 1

CF	Name	Surname	City of Birth
..	..	..	..
..	..	..	..

Table 2

Name	Surname	Date of Birth
..	..	..
..	..	..
..	..	..
..	..	..



Matching
tuples

Not matching

Don't know

RECORD LINKAGE: INPUT OUTPUT DESCRIPTION

Table 1

#	A	C	F
1.1	..	..	..
1.2	..	..	..

Table 2

#	B	C	F
2.1	..	..	..
2.2	..	..	..
2.3	..	..	..
2.4	..	..	..

Record linkage

Matching
tuples

(1.1,2.1,2.3), (1.2,2.2)

Not matching

(1.1,2.2), (1.2,2.3)

Don't know

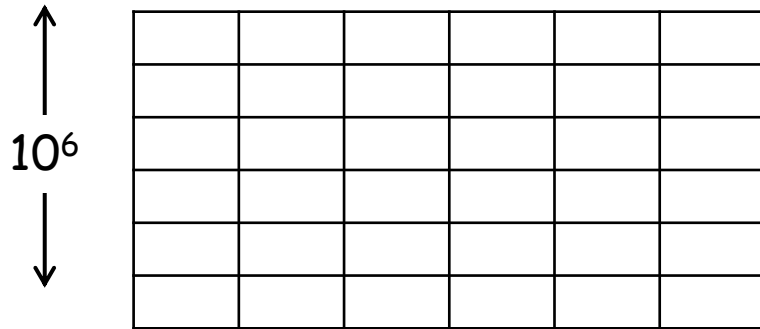
(1.1,2.4),

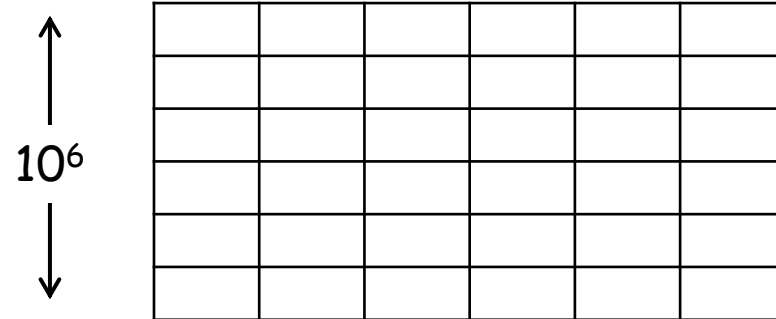
EXISTING TECHNIQUES

- Empirical e.g. “Crlo” in tuple 1.i is similar as “Carlo” in tuple 2.j; so, I match them since they are very close in terms of alphabetic symbols
- Probabilistic:
 - assume I know a **sample** of frequencies of “distances” between pairs of tuples
 - I may decide those matching and non matching ones in the **universe** by projecting such knowledge on the sample on the **universe** of pairs (see later)
- Knowledge based – decision based on **rules** that have to be matched by pairs of tuples
- Mixed probabilistic and knowledge based

HOW MANY PAIRS OF TUPLES TO BE COMPARED IF...

...the two tables have 10^6 each?

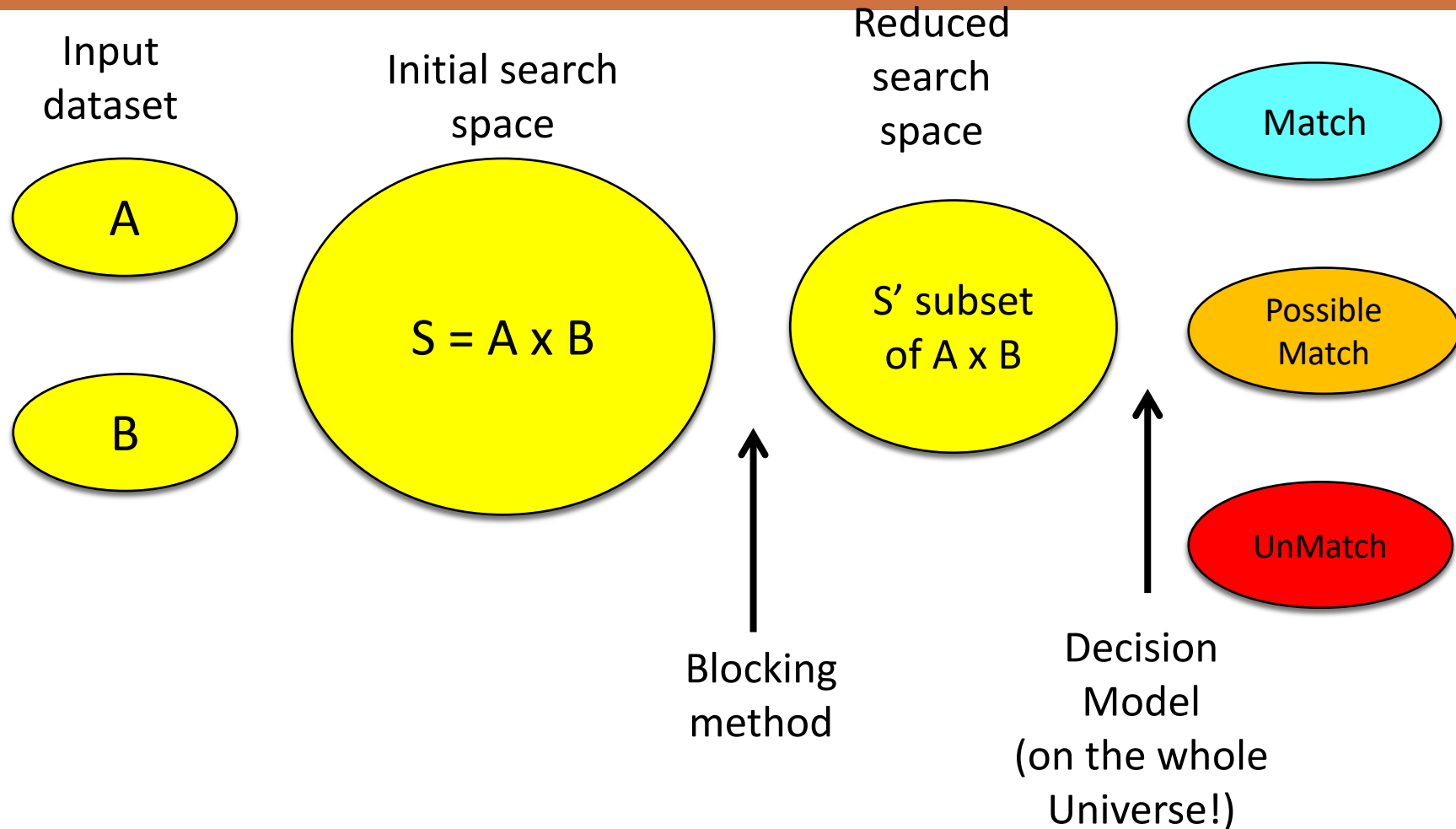




10¹²

STEPS OF RECORD LINKAGE

Assignment



STEPS OF PROBABILISTIC RECORD LINKAGE

- 1. PREPROCESSING - Normalization of formats
- 2. BLOCKING - Blocking with search space reduction
- 3. COMPARE - Choice of distance function
- 4. COMPARE - Find a **sample** of pairs of tuples that are known to be matching (i.e. represent the same observable) or not matching (i.e. represent different observables). Evaluate for each value of distance the frequency of matching and non matching
- 5. DECIDE - Evaluate the distance for pairs in the universe. Analyzing the distribution of step 4, choose two thresholds d_{min} and d_{max}
- 6. DECIDE - For pairs of tuples in the **universe**
 - whose distance d is $d < d_{min} \rightarrow$ decide for matching
 - whose distance d is $d > d_{max} \rightarrow$ decide for non matching
 - whose distance d is $d_{min} < d < d_{max} \rightarrow$ decide for possible match

NORMALIZATION

Re-organized according to a common format

Example

1. Via S. Giovanni 24
2. V. San Giovanni 24
3. V. S. Giovanni 24

Are all transformed into

Via San Giovanni 24

Example

1. 21/12/2021
2. 12/21/2021
3. 13464675326

Are all transformed into

21/12/2021

BLOCKING

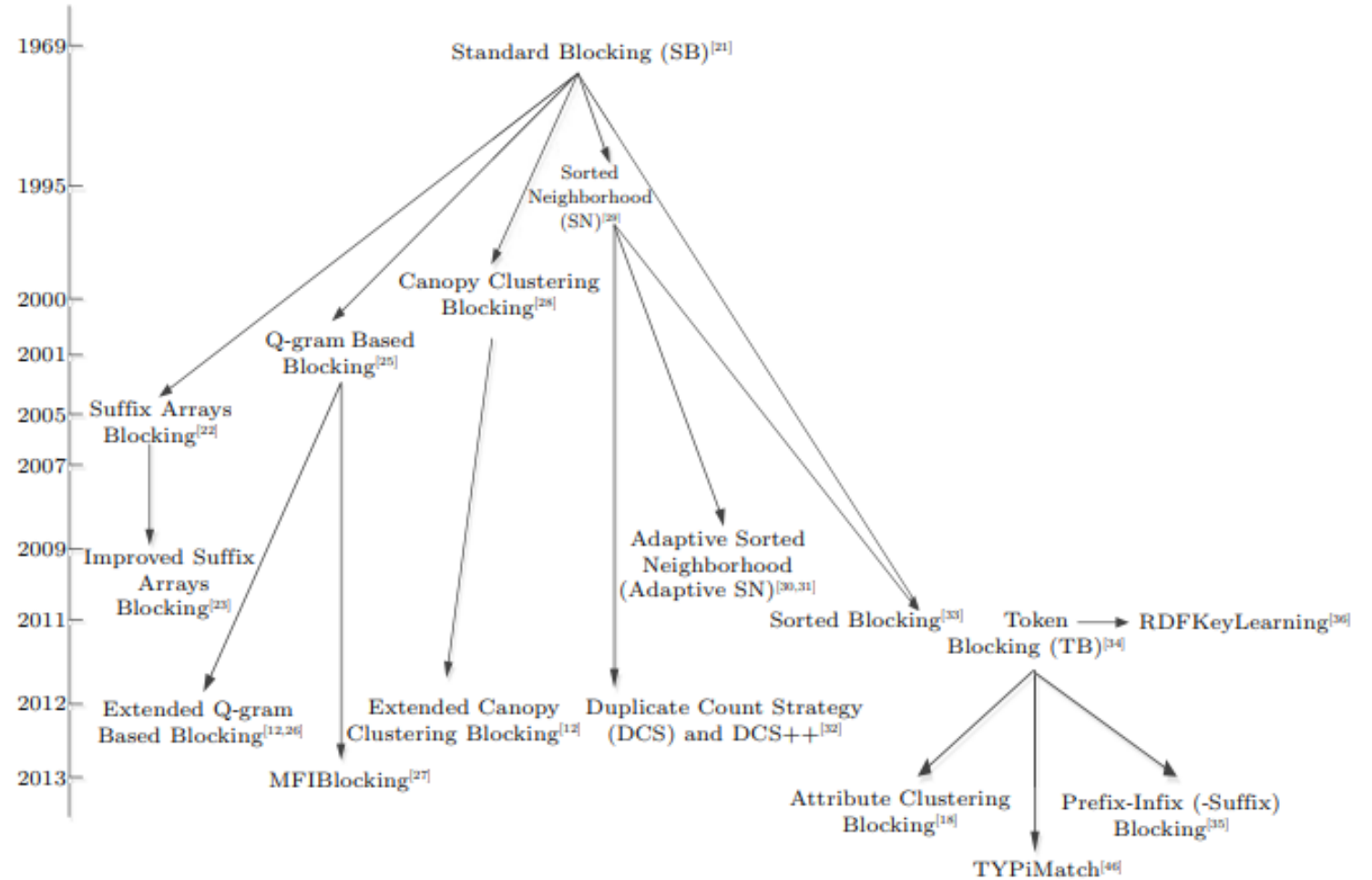
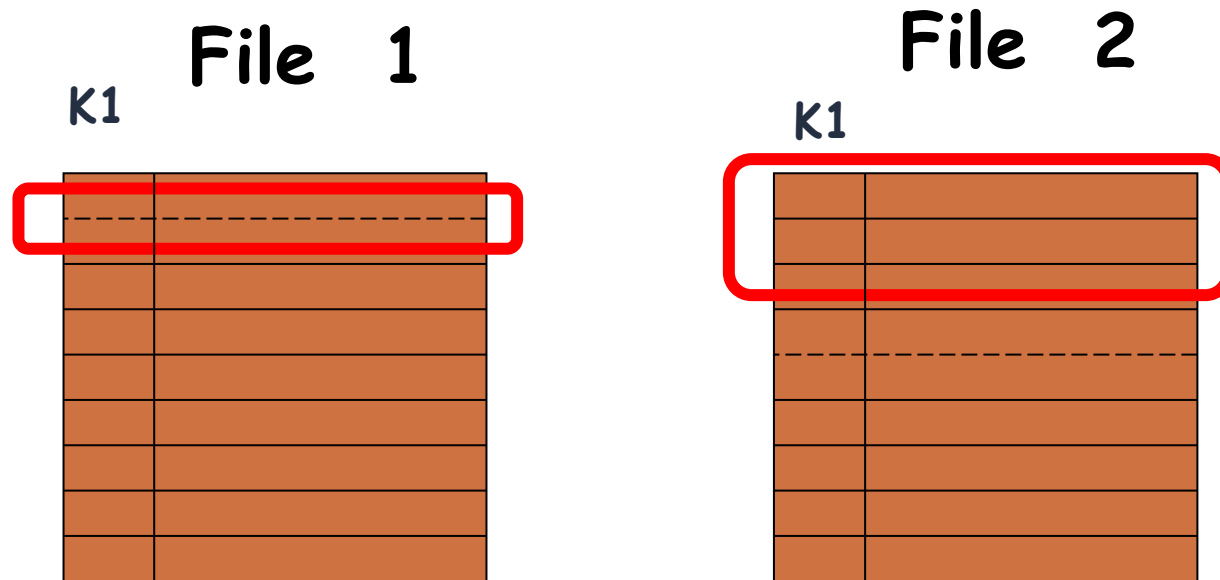


Fig.1. Genealogy trees of block building techniques in chronological order.

SORTED NEIGHBOUR (SN) [HERNANDEZ 1995] STEP I

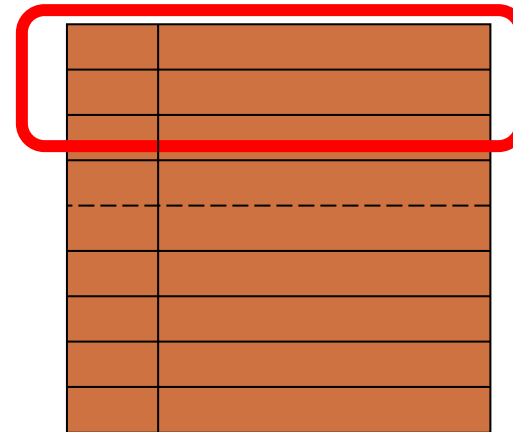
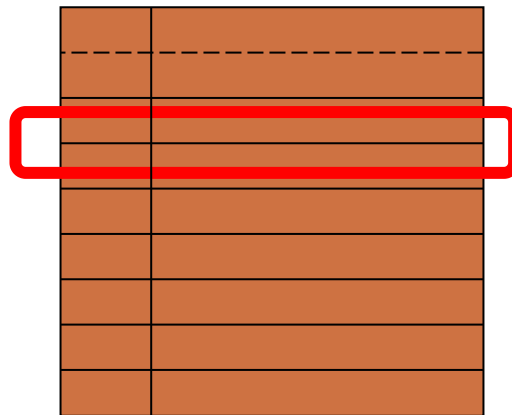
- Implements the blocking method choosing a key, sorting records according to the unique key, and moving a window on sorted files.



SORTED NEIGHBOUR (SN) [HERNANDEZ 1995] STEP 2

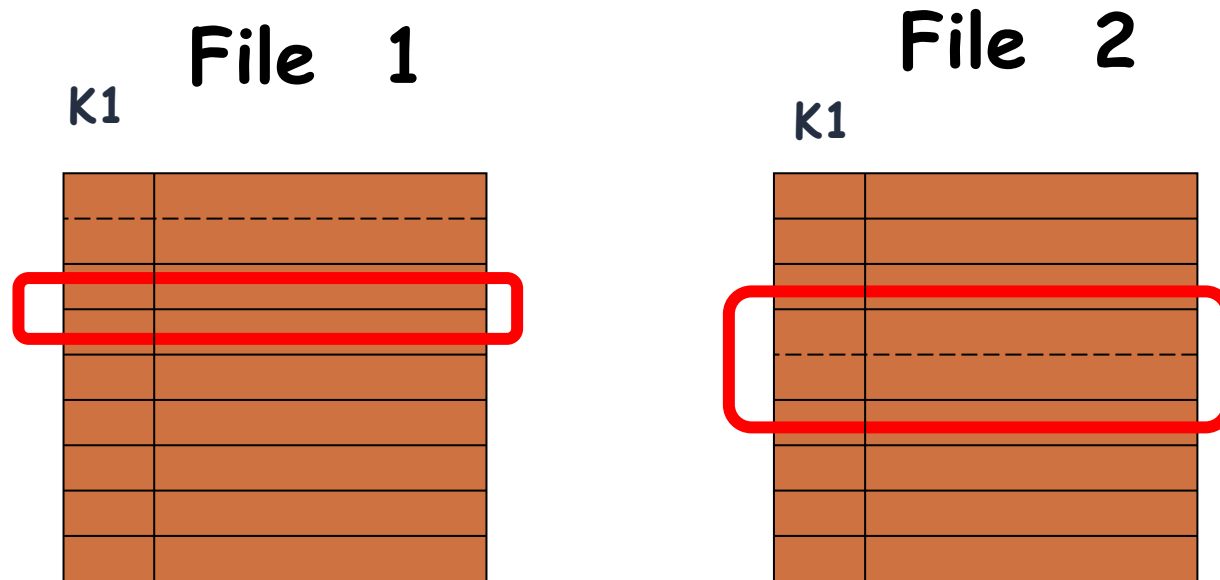


- Implements the blocking method choosing a key, sorting records according to the unique key, and moving a window on sorted files.



SORTED NEIGHBOUR (SN) [HERNANDEZ 1995] STEP 3

- Implements the blocking method choosing a key, sorting records according to the unique key, and moving a window on sorted files.



STEPS OF PROBABILISTIC RECORD LINKAGE

- 1. PREPROCESSING - Normalization of formats
- 2. BLOCKING - Blocking with search space reduction
- 3. COMPARE - Choice of distance function

CHOICE OF DISTANCE FUNCTION

We have to make two choices

- 3.1 Set of attributes involved in the function
- 3.2 Function for distance

STEPS OF PROBABILISTIC RECORD LINKAGE

- 1. PREPROCESSING - Normalization of formats
- 2. BLOCKING - Blocking with search space reduction
- 3. COMPARE - Choice of distance function
- 4. COMPARE - Find a **sample** of pairs of tuples that are known to be matching (i.e. represent the same observable) or not matching (i.e. represent different observables). Evaluate for each value of distance the frequency of matching and non matching

FIND (CHOOSE) THE SAMPLE - 2

Code	Given Name	Last Name	City of birth	Date of birth
034678	Carlo	Batini	Pescara	7/6/1949
543456	Giorgio	Dini	Crema	18/7/1987
543456	Maria	Domi	Milano	28/7/1987

Code	Given Name	Last Name	City of birth	Date of birth
BTNCRL	Crlo	Btini	Pesaro	7/6/1959
DNIGRG	Gioro	Din	Cremona	8/7/1987
DMIGRG	Giorgio	Dami	Modena	8/7/1994
DMISRG	Mria	Domini	Milano	18/9/1994

CHECK IN REAL LIFE...

Code	Given Name	Last Name	City of birth	Date of birth
034678	Carlo	Batini	Pescara	7/6/1949
543456	Giorgio	Dini	Crema	18/7/1987
543456	Maria	Domi	Milano	28/7/1987

Non matching matching

Code	Given Name	Last Name	City of birth	Date of birth
BTNCRL	Crlo	Btini	Pesaro	7/6/1959
DNIGRG	Gioro	Din	Cremona	8/7/1987
DMIGRG	Giorgio	Dami	Modena	8/7/1994
DMISRG	Mria	Domini	Milano	18/9/1994

MATCHING AND NON MATCHING PAIRS IN THE SAMPLE

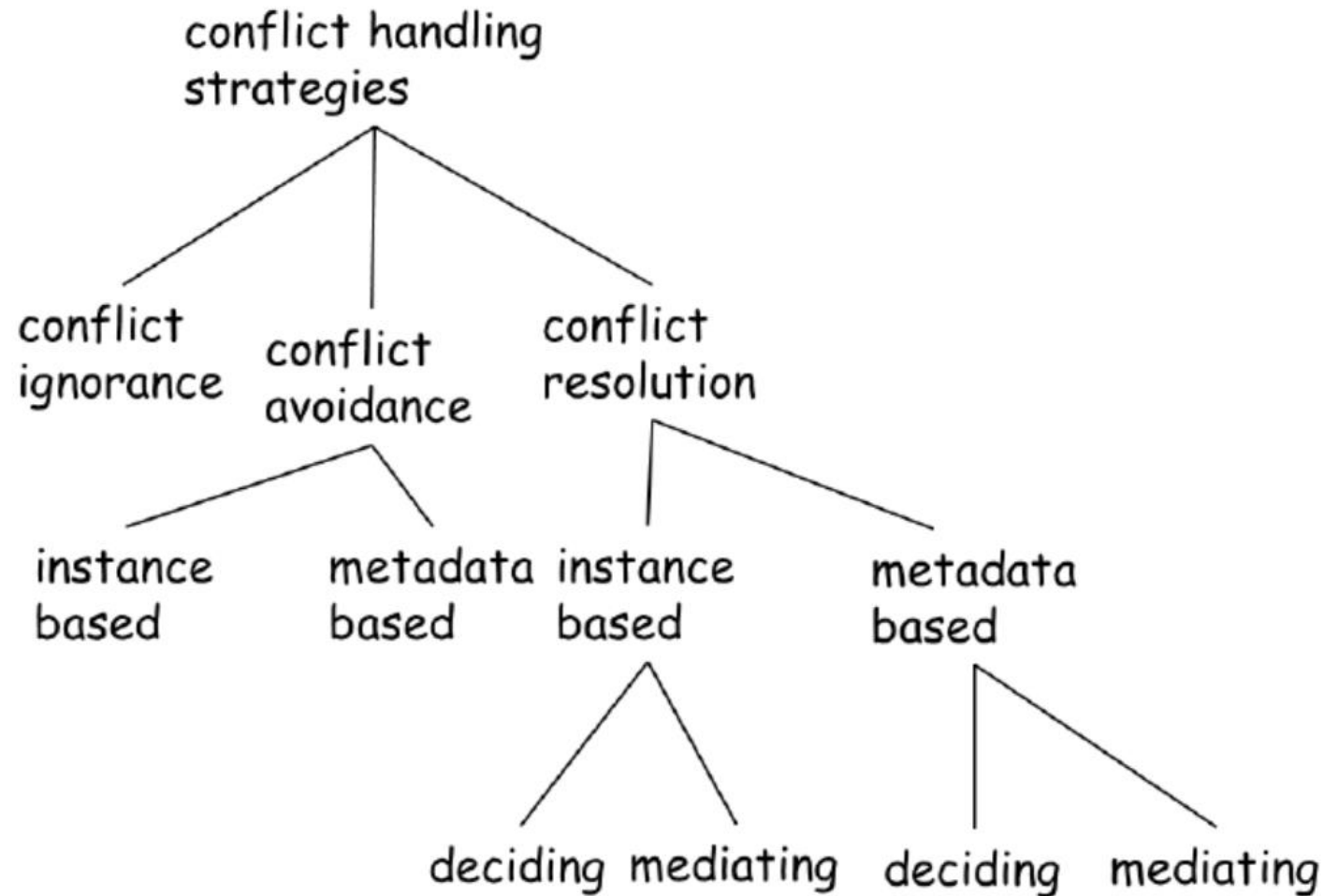
Code	Given Name	Last Name	City of birth	Date of birth
034678	Carlo	Batini	Pescara	7/6/1949
543456	Giorgio	Dini	Crema	18/7/1987
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Code	Given Name	Last Name	City of birth	Date of birth
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STEPS OF PROBABILISTIC RECORD LINKAGE

- 1. PREPROCESSING - Normalization of formats
- 2. BLOCKING - Blocking with search space reduction
- 3. COMPARE - Choice of distance function
- 4. COMPARE - Find a **sample** of pairs of tuples that are known to be matching (i.e. represent the same observable) or not matching (i.e. represent different observables). Evaluate for each value of distance the frequency of matching and non matching
- 5. DECIDE – Evaluate the distance for pairs in the universe. Analyzing the distribution of step 4, choose two thresholds d_{min} and d_{max}
- 6. DECIDE - For pairs of tuples in the universe
 - whose distance d is $d < d_{min} \rightarrow$ decide for matching
 - whose distance d is $d > d_{max} \rightarrow$ decide for non matching
 - whose distance d is $d_{min} < d < d_{max} \rightarrow$ decide for possible match

DATA FUSION (CONFLICT HANDLING) STRATEGIES



POSSIBLE CONFLICT HANDLING STRATEGIES

Strategy	Classification	Short Description
PASS IT ON	ignoring	escalates conflicts to user or application
CONSIDER ALL POSSIBILITIES	ignoring	creates all possible value combinations
TAKE THE INFORMATION	avoiding, instance based	prefers values over null values
NO GOSSIPING	avoiding, instance based	returns only consistent tuples
TRUST YOUR FRIENDS	avoiding, metadata based	takes the value of a preferred source
CRY WITH THE WOLVES	resolution, instance based, deciding	takes the most often occurring value
ROLL THE DICE	resolution, instance based, deciding	takes a random value
MEET IN THE MIDDLE	resolution, instance based, mediating	takes an average value
KEEP UP TO DATE	resolution, metadata based, deciding	takes the most recent value

CONFLICT IGNORING

Conflict-ignoring strategies do not make a decision as to what to do with conflicting data and sometimes are not even aware of data conflicts. An example for an ignoring strategy is Pass It On, which presents all values and that way defers conflict resolution to the user.

Strategy	Classification	Short Description
PASS IT ON	ignoring	escalates conflicts to user or application
CONSIDER ALL POSSIBILITIES	ignoring	creates all possible value combinations
TAKE THE INFORMATION	avoiding, instance based	prefers values over null values
NO GOSSIPING	avoiding, instance based	returns only consistent tuples
TRUST YOUR FRIENDS	avoiding, metadata based	takes the value of a preferred source
CRY WITH THE WOLVES	resolution, instance based, deciding	takes the most often occurring value
ROLL THE DICE	resolution, instance based, deciding	takes a random value
MEET IN THE MIDDLE	resolution, instance based, mediating	takes an average value
KEEP UP TO DATE	resolution, metadata based, deciding	takes the most recent value

City	Source: Milan Municipality	Milano
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City	Source: INPS	Roma
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City	Source: Milan Municipality	Milano	Source: INPS	Roma
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CONFLICT-AVOIDING STRATEGIES

Conflict-avoiding strategies acknowledge the existence of possible conflicts in general, but do not detect and resolve single existing conflicts. Instead, they handle conflicting data by applying **a unique decision equally to all data**, such as preferring data from a special source with the Trust Your Friends strategy.

Strategy	Classification	Short Description
PASS IT ON	ignoring	escalates conflicts to user or application
CONSIDER ALL POSSIBILITIES	ignoring	creates all possible value combinations
TAKE THE INFORMATION	avoiding, instance based	prefers values over null values
NO GOSSIPING	avoiding, instance based	returns only consistent tuples
TRUST YOUR FRIENDS	avoiding, metadata based	takes the value of a preferred source
CRY WITH THE WOLVES	resolution, instance based, deciding	takes the most often occurring value
ROLL THE DICE	resolution, instance based, deciding	takes a random value
MEET IN THE MIDDLE	resolution, instance based, mediating	takes an average value
KEEP UP TO DATE	resolution, metadata based, deciding	takes the most recent value

City	Source: Milan Municipality	Milano
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City	Source: INPS	Roma
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City	Source: Milan Municipality	Milano
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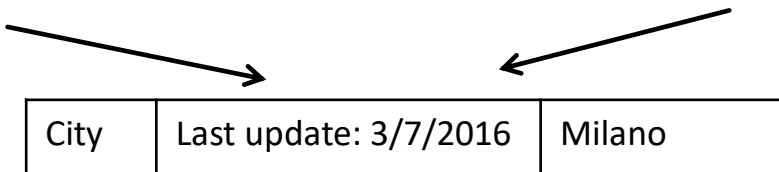
CONFLICT RESOLUTION STRATEGIES

Conflict *resolution* strategies do **regard all the data and metadata** before deciding on how to resolve a conflict. They can further be subdivided into *deciding* when they choose a value from all the already present values. E.g. assuming that timestamp information is given for all values, applying strategy Keep Up To Date takes the most recent value.

Strategy	Classification	Short Description
PASS IT ON	ignoring	escalates conflicts to user or application
CONSIDER ALL POSSIBILITIES	ignoring	creates all possible value combinations
TAKE THE INFORMATION	avoiding, instance based	prefers values over null values
NO GOSSIPING	avoiding, instance based	returns only consistent tuples
TRUST YOUR FRIENDS	avoiding, metadata based	takes the value of a preferred source
CRY WITH THE WOLVES	resolution, instance based, deciding	takes the most often occurring value
ROLL THE DICE	resolution, instance based, deciding	takes a random value
MEET IN THE MIDDLE	resolution, instance based, mediating	takes an average value
KEEP UP TO DATE	resolution, metadata based, deciding	takes the most recent value

City	Last update: 3/7/2016	Milano
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City	Last update: 3/9/2014	Roma
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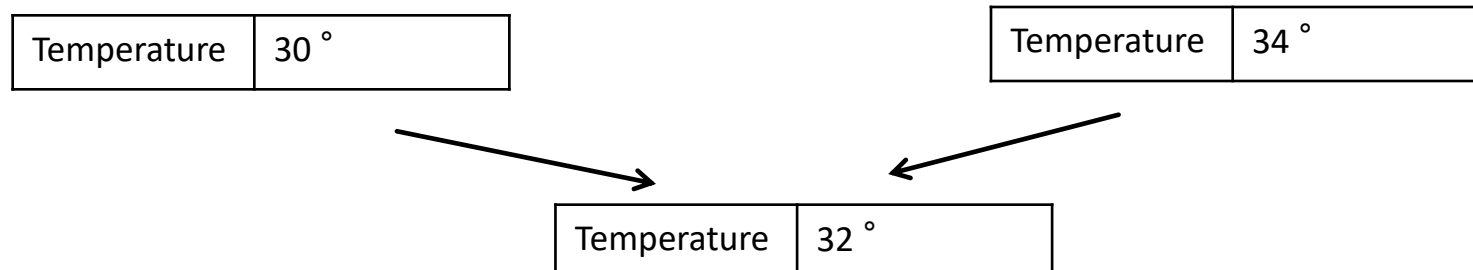


City	Last update: 3/7/2016	Milano
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CONFLICT *RESOLUTION* STRATEGIES

Conflict *resolution* strategies do **regard all the data and metadata** before deciding on how to resolve a conflict. They can further be subdivided into

- *deciding* when they choose a value from all the already present values. E.g. assuming that timestamp information is given for all values, applying strategy Keep Up To Date takes the most recent value.
- *mediating* strategies, when they choose a value that does not necessarily exist among the conflicting values (mediating).



RECORD LINKAGE OF TWO TABLES

RULE OF THUMB: RECORDS ARE THE SAME IF **THEY DIFFER ONLY IN ONE LETTER IN NAME & SURNAME**

Name	Surname	Place of birth	CountryB
Crlo	Batini	Pescara	Italy
Giulio	Sandri	Milano	Italy
Leo	Siti	Milano	Italy
Gianni	Batini	Paris	null

Name	Surname	Place of birth	Date of B.
Carlo	Batini	Pescara	3-6-2007
null	Sandri	Roma	4-7-1998
Leonardo	Siti	Milano	24-7-2011
Gianni	null	Paris	13-3-1955

Rule of thumb: records are the same if **they differ only in one letter in Name & Surname**

RECORD LINKAGE OF TWO TABLES

Name	Surname	Place of birth	CountryB
Crlo	Batini	Pescara	Italy
Giulio	Sandri	Milano	Italy
Leo	Siti	Milano	Italy
Gianni	Batini	Paris	null

Name	Surname	Place of birth	Date of B.
Carlo	Batini	Pescara	3-6-2007
null	Sandri	Roma	4-7-1998
Leonardo	Siti	Milano	24-7-2011
Gianni	null	Paris	13-3-1955

Rule of thumb: records are the same if they differ only in one letter in Name & Surname **or** the name in one table is a known contraction of the name in the second one

RECORD LINKAGE OF TWO TABLES

Name	Surname	Place of birth	Country B
Crlo	Batini	Pescara	Italy
Giulio	Sandri	Milano	Italy
Leo	Siti	Milano	Italy
Gianni	Batini	Paris	null

Name	Surnam	Place of birth	Date of B
Carlo	Batini	Pescara	3-6-2007
null	Sandri	Roma	4-7-1998
Leonardo	Siti	Milano	24-7-2011
Gianni	null	Paris	13-3-1955

Rule of thumb: records are the same if they differ only in one letter in Name & Surname **or the name in one table is a known contraction of the name in the second one and the place of birth is the same**

Name	Surname	Place of birth	Date of B.	Country B
Carlo	Batini	Pescara	3-6-2007	Italy
Giulio	Sandri	Milano	null	Italy
null	Sandri	Roma	4-7-1998	null
Leonardo	Siti	Milano	24-7-2011	Italy
Gianni	Batini	Paris	null	null
Gianni	null	Paris	13-3-1955	null

IMPROVEMENT ON COUNTRYB

Name	Surname	Place of birth	Country B
Crlo	Batini	Pescara	Italy
Giulio	Sandri	Milano	Italy
Leo	Siti	null	Italy
Gianni	Batini	Paris	null

Name	Surnam e	Place of birth	Date of B.
Carlo	Batini	Pescara	3-6-2007
Giulio	Sandri	Roma	4-7-1998
Leonardo	Siti	Milano	24-7-2011
Gianni	null	Paris	13-3-1955

Name	Surname	Place of birth	Date of B.	Country B
Carlo	Batini	Pescara	3-6-2007	Italy
Giulio	Sandri	Milano	null	Italy
null	Sandri	Roma	4-7-1998	null
Leonardo	Siti	Milano	24-7-2011	Italy
Gianni	Batini	Paris	null	null
Gianni	null	Paris	13-3-1955	null

FINAL RESULT

Name	Surname	Place of birth	Country B
Crlo	Batini	Pescara	Italy
Giulio	Sandri	Milano	Italy
Leo	Siti	null	Italy
Gianni	Batini	Paris	null

Name	Surnam e	Place of birth	Date of B.
Carlo	Batini	Pescara	3-6-2007
Giulio	Sandri	Roma	4-7-1998
Leonardo	Siti	Milano	24-7-2011
Gianni	null	Paris	13-3-1955

Name	Surname	Place of birth	Date of B.	Country B
Carlo	Batini	Pescara	3-6-2007	Italy
Giulio	Sandri	Milano	null	Italy
null	Sandri	Roma	4-7-1998	Italy
Leonardo	Siti	Milano	24-7-2011	Italy
Gianni	Batini	Paris	null	France
Gianni	null	Paris	13-3-1955	France